

Environmental Kuznets Curve Analysis for BRICS Countries

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Contents

Abstract	2
Introduction	2
1 Hypothesis of EKC and Literature Review	2
1.1 EKC Theory	2
1.2 Different Specifications of the EKC	3
2 Data and Methodology	5
2.1 Literature on BRICS	5
2.2 Data Sources and Variables	6
3 Results and Discussion	6
3.1 EKC Shape and Turning Point	6
3.2 Differences across BRICS	10
3.3 Conclusion	14
References	15

Abstract

This paper examines the relationship between economic growth and environmental sustainability in the BRICS countries, with particular attention to the People’s Republic of China, by empirically estimating Environmental Kuznets Curve (EKC) models using panel data from 1988 to 2023. This survey categorizes the studies based on single-country and cross-country contexts. The results indicate that, for the BRICS as a whole, there is no evidence of an inverted U-shape relationship; however, when the countries are considered individually, the scenario differs. The data are processed solely as a function of GDP and per capita CO₂ emissions. Other factors that may influence the EKC are intentionally excluded to isolate the most direct effect—namely, that of GDP. The Environmental Kuznets Curve is modeled using both second- and third-degree polynomial functions.

Introduction

At the onset of economic growth, nations often experience a significant decline in environmental quality. This initial stage of development is typically characterized by heightened levels of ambient air pollution, deforestation, and contamination of soil and water resources. As income levels increase and the economy advances, the rate of environmental degradation tends to decelerate. Eventually, upon reaching a certain income threshold, the trend reverses—environmental degradation diminishes, and environmental quality begins to improve.

This hypothesized non-linear relationship between economic growth and carbon dioxide (CO₂) emissions is commonly represented by an inverted U-shaped curve, known as the Environmental Kuznets Curve (EKC) hypothesis. The term derives from the work of Simon Kuznets (1955), who originally identified a similar inverted U-shaped relationship between economic growth and income inequality. Subsequently, Grossman and Krueger (1995) observed a comparable pattern in the context of economic growth and environmental degradation, thereby extending the conceptual framework to environmental economics.

1 Hypothesis of EKC and Literature Review

1.1 EKC Theory

The Environmental Kuznets Curve (EKC) hypothesis posits a non-linear relationship between economic growth and environmental degradation, typically taking the form of an inverted U-shape. At the early stages of development, as GDP rises, environmental quality tends to decline due to intensified resource extraction and pollution-intensive

industrialization. This phase is largely driven by the scale effect, wherein economic expansion increases the demand for natural resources and leads to greater emissions and waste.

As income grows, however, the economic structure begins to evolve. The composition effect emerges as economies transition from resource-intensive sectors (agriculture and manufacturing) toward services and cleaner industries. This structural shift is often accompanied by urbanization and increasing public demand for environmental protection.

In more advanced stages, the technique effect becomes predominant. Technological innovation, energy efficiency improvements, and stricter environmental regulations contribute to a decoupling of growth and pollution. Firms adopt greener technologies, and governments reinforce environmental standards, often in response to rising income elasticity of environmental demand—the idea that demand for environmental quality increases with income.

This dynamic helps explain the observed turning point in environmental degradation, marking the start of environmental improvements despite continued economic growth. From a market-oriented perspective, increasing scarcity of natural resources leads to higher prices, which can incentivize firms to adopt resource-efficient technologies (Duflou et al., 2012). Thus, price signals and endogenous market adjustments may also contribute to the declining phase of the EKC.

In sum, the EKC hypothesis reflects the interplay between economic incentives, structural transformation, technological change, and policy responses, all of which evolve endogenously with income.

1.2 Different Specifications of the EKC

Although the empirical literature on the estimation of the Environmental Kuznets Curve (EKC) for CO₂ emissions is extensive, many studies share a common modeling framework. Most contributions employ either cross-sectional or panel data techniques and rely on polynomial regressions of per capita income to capture the non-linear relationship between economic growth and environmental degradation.

The general form of the EKC model used in the literature is as follows:

$$\text{CO}_{2,pc} = \alpha + \beta_1 \text{GDP}_{pc} + \beta_2 \text{GDP}_{pc}^2 + \beta_3 \text{GDP}_{pc}^3 + \varepsilon_{it} \quad (1)$$

where:

- CO₂ denotes per capita CO₂ emissions for country i at time t ,
- GDP_{pc} is per capita income (GDP),
- α is the intercept,
- β_k are coefficients associated with the income terms,

- ε_{it} is the error term.

Most of the time, however, it is preferable to use a logarithmic specification for ease of calculation and for correct data processing. This flexible specification allows for several functional forms of the income–emissions relationship:

- (a) $\beta_1 = \beta_2 = \beta_3 = 0$: No relationship between income and emissions.
- (b) $\beta_1 > 0, \beta_2 = \beta_3 = 0$: Linear and increasing relationship.
- (c) $\beta_1 < 0, \beta_2 = \beta_3 = 0$: Linear and decreasing relationship.
- (d) $\beta_1 > 0, \beta_2 < 0, \beta_3 = 0$: Inverted U-shaped curve (standard EKC).
- (e) $\beta_1 < 0, \beta_2 > 0, \beta_3 = 0$: U-shaped relationship.
- (f) $\beta_1 > 0, \beta_2 < 0, \beta_3 > 0$: N-shaped relationship.
- (g) $\beta_1 < 0, \beta_2 > 0, \beta_3 < 0$: Inverted N-shaped relationship.

The most widely accepted form consistent with the EKC hypothesis is case (d), where an inverted U-shape suggests that emissions initially rise with income, reach a peak, and then decline as income continues to grow. The income level at which emissions peak (the turning point) in the quadratic model is given by:

$$GDP^* = -\frac{\beta_1}{2\beta_2} \quad (2)$$

In the cubic (N-shaped) specification, the turning points are obtained by solving the derivative of equation (1) with respect to income:

$$GDP^* = \frac{-\beta_2 \pm \sqrt{\beta_2^2 - 3\beta_1\beta_3}}{3\beta_3} \quad (3)$$

Higher-order polynomials enable the model to capture more complex nonlinearities and assess the long-run sustainability of economic growth. In this context, the signs and magnitudes of β_1, β_2 , and β_3 determine not only the shape of the EKC but also the position and stability of the turning points. Therefore, understanding the functional form is crucial for accurate policy implications regarding decarbonization and sustainable development.

2 Data and Methodology

2.1 Literature on BRICS

The acronym BRIC was coined in 2001 by Goldman Sachs economist Jim O’Neill ¹ to designate four emerging economies—Brazil, Russia, India, and China—shared by high growth rates and a potential systemic influence on the global economy. Although initially formulated in analytical and forecasting terms, starting in 2006 these countries initiated a multilateral political dialogue, which culminated in the first official summit in Ekaterinburg in 2009. South Africa’s accession in 2013, transforming the acronym into BRICS, also gave the group continental geopolitical legitimacy. Since then, the BRICS have sought to position themselves as an alternative to Western-dominated global power structures, through institutions such as the New Development Bank (2014) and the Contingent Reserves Arrangement, designed as financing instruments parallel to the IMF and the World Bank.

However, behind this shared rhetoric, profound differences emerge. China is the dominant systemic player: a manufacturing power and creditor, it relies on a strong state apparatus and targeted foreign investment policies, particularly along the Belt and Road (e.g., Sudan, Pakistan, Kazakhstan). India, while sharing the same demographic weight and role as a technology hub, adopts a different strategy, focusing on bureaucratic simplification to attract foreign capital while maintaining a democratic system. Russia emerges as a revisionist energy power, with an assertive geopolitical approach and on a collision course with the liberal order. Brazil, though democratic, suffers from recurrent political instability and heavy dependence on raw materials. Finally, South Africa plays a predominantly symbolic role, representing sub-Saharan Africa.

Divergences among members are manifested in political models, productive bases (industry, services, energy), and strategic priorities (multipolarity, de-dollarization, development). Rather than a cohesive bloc, the BRICS represent a functional and pragmatic alliance, oriented toward a shared desire to reform the global order, yet profoundly marked by internal asymmetries and divergent geopolitical visions ².

¹<https://www.goldmansachs.com/insights/goldman-sachs-research/building-better>

²<https://www.cfr.org/background/what-brics-group-and-why-it-expanding>

2.2 Data Sources and Variables

The data for carbon dioxide (CO₂) emissions excluding LULUCF³ per capita (tCO₂e/capita) come from EDGAR (Emissions Database for Global Atmospheric Research)⁴, while for GDP per capita (current US\$) it comes from the World Bank⁵. The data were extrapolated and cleaned from 1988 to 2023 in order to have a period long enough (36 years) to include economic booms, the birth of the internet, and political initiatives to promote the decoupling of emissions and production.

3 Results and Discussion

3.1 EKC Shape and Turning Point

As reported in the first part, I adopted two functional specifications to estimate the Environmental Kuznets Curve: a second-degree polynomial and a third-degree polynomial. This choice allows us to identify any nonlinearities or complex behaviors that a single specification might miss. The curve obtained with the first specification is as follows:

$$\ln(\text{CO}_{2,pc}) = \alpha + \beta_1 \ln(\text{GDP}_{pc}) + \beta_2 [\ln(\text{GDP}_{pc})]^2 + \varepsilon_{it}$$

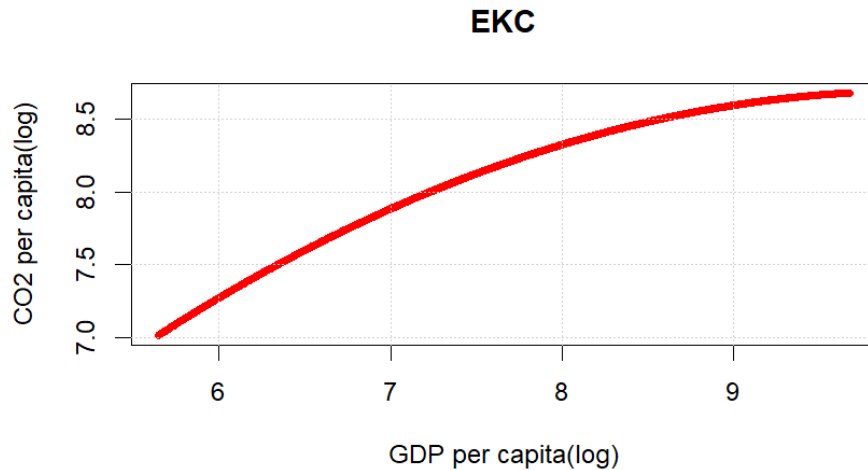


Figure 1: Chart reports the effect of the (log of) GDP per capita on the (log of) emissions, for different levels of the (log of) GDP per capita. Estimate is based on BRICS countries, over the period 1988–2023. *Twoways effects Within Model*

As can be seen from a simple graphical analysis, the BRICS countries taken together are not yet on the path to decoupling carbon dioxide emissions from production growth.

³Land Use, Land-Use Change and Forestry

⁴Further analysis can be done here: <https://edgar.jrc.ec.europa.eu/>

⁵More informations can be found at: <https://data.worldbank.org/>

Coefficients	Estimate	Std. Error	t-value	Pr(> t)
$\ln GDP$	1.7257581	0.1279604	13.487	< 2.2e-16
$\ln GDP^2$	-0.0856410	0.0084345	-10.154	< 2.2e-16

Table 1: Coefficient Estimates and Statistics

This is likely due to their persistent dependence on fossil fuels (consider that China still relies 61% on coal ⁶) and the lack of cooperation between countries, or rather, the delay in adopting policies to reduce carbon dioxide emissions and other pollutants.

To identify the income level at which carbon dioxide emissions begin to decline, i.e. the turning point of the Environmental Kuznets Curve, we apply the standard formula for a log-log quadratic specification:

$$\ln(GDP_{pc})^* = -\frac{\beta_1}{2\beta_2} \Rightarrow GDP_{pc}^* = e^{-\beta_1/(2\beta_2)}$$

Using the estimated coefficients from the second-degree polynomial regression:

$$\beta_1 = 1.7257581 \quad \text{and} \quad \beta_2 = -0.0856410$$

The turning point in logarithmic terms is:

$$\ln(GDP_{pc})^* = -\frac{1.7257581}{2 \times (-0.0856410)} \approx 10.075$$

Exponentiating both sides gives:

$$GDP_{pc}^* = e^{10.075} \approx 23741.97$$

This implies that the emissions begin to decline when GDP per capita reaches approximately \$23,697. Below this level, economic growth is associated with rising emissions, obviously due to production's reliance on fossil fuels to meet the ever-increasing demand for goods and services and to support emerging economies of scale (e.g., China, India, Russia). Above this level, further growth is linked to a reduction in emissions following investments in research and development (R&D) for the use of renewable energy and zero-impact factories, in line with the hypothesized inverted U-shape of the EKC. It should be emphasized, however, that this turning point is only a theoretical number at which the decoupling of emissions from production should begin in the long run.

⁶<https://www.iea.org/countries/china/energy-mix>

With the second specification, i.e. the third-degree polynomial regression, the following results are obtained:

$$\ln(\text{CO}_{2,pc}) = \alpha + \beta_1 \ln(\text{GDP}_{pc}) + \beta_2 [\ln(\text{GDP}_{pc})]^2 + \beta_3 [\ln(\text{GDP}_{pc})]^3 + \varepsilon_{it}$$

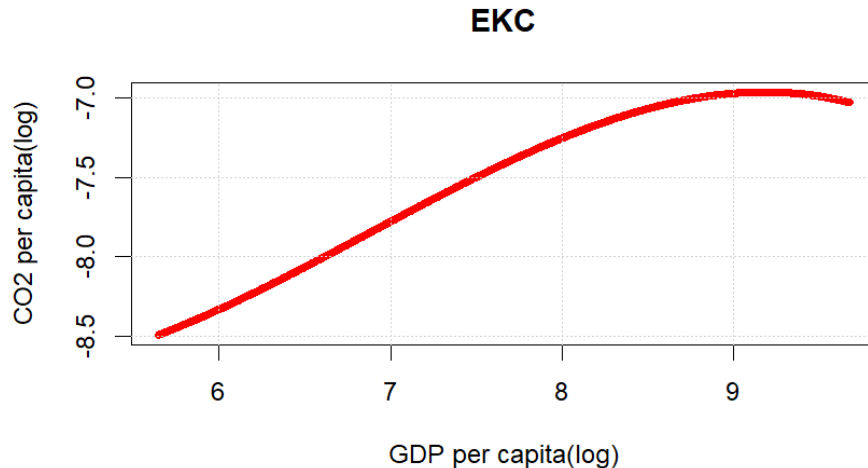


Figure 2: Chart reports the effect of the (log of) GDP per capita on the (log of) emissions, for different levels of the (log of) GDP per capita. Estimate is always based on BRICS countries, over the period 1988–2023. *Twoways effects Within Model*

	Estimate	Std. Error	t-value	$Pr(> t)$
$\ln GDP$	-4.5836602	1.6963044	-2.7021	0.0077608
$\ln GDP^2$	0.7508910	0.2244630	3.3453	0.0010608
$\ln GDP^3$	-0.0364024	0.0097614	-3.7292	0.0002803

Table 2: Coefficient Estimates and Statistics

Starting from the assumption that the coefficients are statistically significant, with this specification, the result does not change remarkably, since β_2 is positive and close to 1, while β_3 is negative and close to zero. As indicated in formulas (c) and (g), we expect, in the long run, either a linear and decreasing relationship or an inverted N-shaped curve. The actual shape will depend on the environmental policies implemented and the ability to optimize production from an energy-efficiency perspective. Consider that the BRICS countries have only recently begun to adopt policies to decarbonize industry and protect the environment. For example, Brazil, under President Lula, reactivated the Action Plan to Prevent and Control Deforestation in the Legal Amazon ⁷ by 2030. Furthermore, it is increasingly consuming energy from renewable sources, including renewable biofuels, mostly plant-based, and waste products, such as household and industrial waste, which can be burned to generate electricity or heat, in fact, Brazil consumed 43% of its final energy consumption from modern renewables ⁸.

⁷<https://infoamazonia.org/>

⁸<https://www.iea.org/regions/central-south-america/renewables>

China, for its part, has set ambitious but complex goals for the ecological transition. President Xi Jinping announced in 2020 that the country will reach peak emissions by 2030 and carbon neutrality by 2060 ⁹. To achieve these goals, a climate governance system known as “1+N” was launched, which includes a national framework plan accompanied by specific sectoral plans for energy, transportation, real estate, and agriculture ¹⁰.

From an industrial perspective, China is now a world leader in renewable energy, especially wind and photovoltaic power, and is investing heavily in the electrification of transportation (electric cars, high-speed rail networks). However, it remains heavily dependent on coal, which, as stated at the outset, still accounted for approximately 60% of the national energy mix in 2024. Internationally, China has strengthened its presence at climate conferences (COPs) and increased its financing of green projects, including in partner countries along the Belt and Road Initiative. Nevertheless, by analyzing the level of GDP per capita at the turning point to understand at what level of product per person the hoped-for inversion will begin, with the specific formula for the third degree polynomial (3), we obtain:

$$\frac{d \ln(\text{CO}_{2,pc})}{d \ln(\text{GDP}_{pc})} = \beta_1 + 2\beta_2 \ln(\text{GDP}_{pc}) + 3\beta_3 [\ln(\text{GDP}_{pc})]^2 = 0$$

Let us set $\ln(\text{GDP}_{pc}) = x$, and substitute the estimated coefficients into the equation:

$$-4.58 + 2(0.75) \cdot x + 3(-0.036) \cdot x^2 = 0$$

$$-0.11x^2 + 1.50x - 4.58 = 0$$

Solving using the quadratic formula:

$$x_{1,2} = \frac{-1.50 \pm \sqrt{1.50^2 - 4 \cdot (-0.11) \cdot (-4.58)}}{2 \cdot (-0.11)}$$

$$x = \frac{-1.50 \pm \sqrt{0.2348}}{-0.22} \Rightarrow x_1 \approx 9.02, \quad x_2 \approx 4.61$$

$$\text{GDP}_{pc} \approx e^{9.02} \approx 8266.77, \quad \text{and} \quad e^{4.61} \approx 100.48$$

The presence of two turning points between \$100 and \$8266.77 suggests an inverted-N shaped EKC. Initially, emissions increase with income, then decrease, and may eventually rise again at higher income levels, depending on economic structure and technological transitions.

⁹<https://www.eeas.europa.eu>

¹⁰<https://www.undp.org/china>

3.2 Differences across BRICS

A closer look at the BRICS countries allows us to better analyze their differences and similarities. As mentioned, I used only the most common formulas to test the EKC (second- and third-degree polynomials) without including dummy variables such as agricultural policies, population growth, land use, renewable energy consumption, and the impact of free trade. Below are the graphs and coefficients obtained by regressing only per capita carbon dioxide emissions with the squares of per capita income:

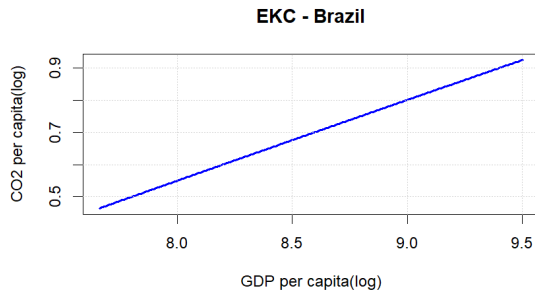


Table 3: Regression Coefficients – Brazil

Coeff.	Estimate	Std. Err.	t-val	Pr(> t)
Intercept	-1.592	3.433	-0.464	0.646
$\ln GDP$	0.283	0.803	0.353	0.727
$\ln GDP^2$	-0.0019	0.047	-0.041	0.967

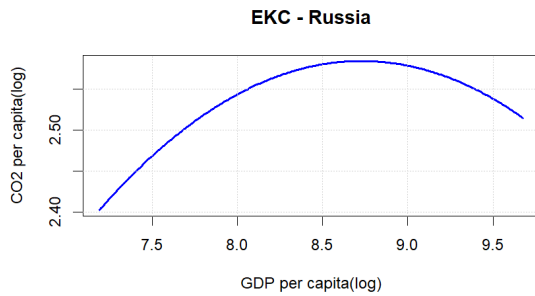


Table 4: Regression Coefficients – Russia

Coeff.	Estimate	Std. Err.	t-val	Pr(> t)
Intercept	-3.282	2.948	-1.113	0.274
$\ln GDP$	1.344	0.691	1.945	0.060
$\ln GDP^2$	-0.077	0.040	-1.914	0.064

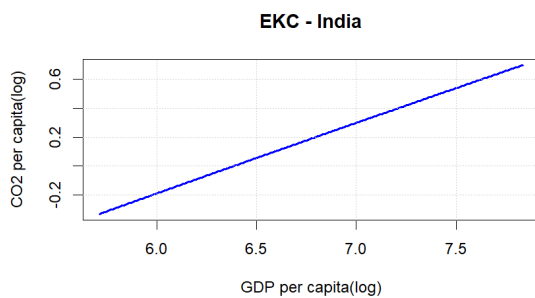


Table 5: Regression Coefficients – India

Coeff.	Estimate	Std. Err.	t-val	Pr(> t)
Intercept	-3.334	1.520	-2.194	0.035
$\ln GDP$	0.554	0.456	1.215	0.233
$\ln GDP^2$	-0.005	0.034	-0.148	0.883

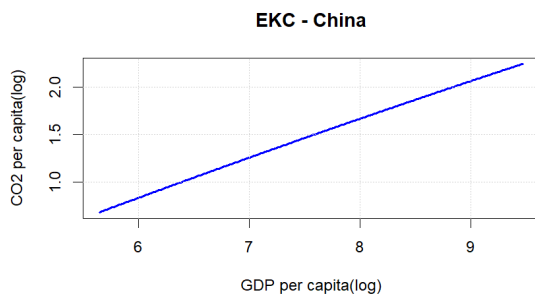


Table 6: Regression Coefficients – China

Coeff.	Estimate	Std. Err.	t-val	Pr(> t)
Intercept	-1.988	0.594	-3.344	0.002
$\ln GDP$	0.511	0.160	3.203	0.003
$\ln GDP^2$	-0.0067	0.010	-0.645	0.523

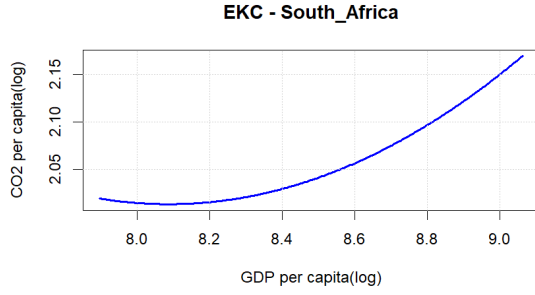


Table 7: Regression Coefficients – South Africa

Coeff.	Estimate	Std. Err.	t-val	Pr(> t)
Intercept	12.700	11.450	1.109	0.275
$\ln GDP$	-2.643	2.716	-0.973	0.337
$\ln GDP^2$	0.163	0.161	1.017	0.317

As we can see, only Russia exhibits a typical environmental Kuznets curve (the inverted U shape). Brazil, India, China, and South Africa appear to have not seen a decrease in CO₂ emissions and overall appear to have a positive linear increasing trend and are therefore very far from the decoupling between per capita carbon dioxide emissions and the increase in per capita production. That said, I calculated the GDP per capita level only for Russia, which turns out to be \$6168,88.

As already mentioned, to implement information that would otherwise not be demonstrable with a regression on CO₂ per capita explained only by the square of the GDP per capita, I also used a third-degree polynomial function to understand the effects of a substantial increase in per capita income in the long run and to understand whether the countries in question are capable of obtaining shapes equal to the U shape (inversed) or the N shape (inversed). Below are the results obtained:

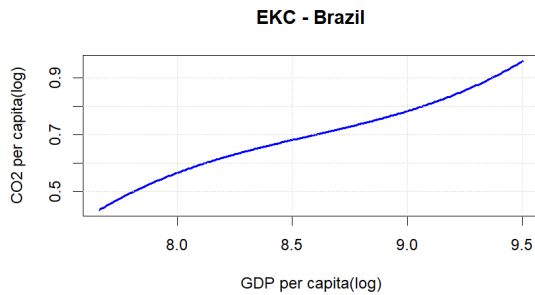


Table 8: Regression Coefficients – Brazil

Coeff.	Estimate	Std. Err.	t-val	Pr(> t)
Intercept	-74.81806	55.82019	-1.340	0.190
$\ln GDP$	26.03040	19.60682	1.328	0.194
$\ln GDP^2$	-3.01195	2.29076	-1.315	0.198
$\ln GDP^3$	0.11700	0.08903	1.314	0.198

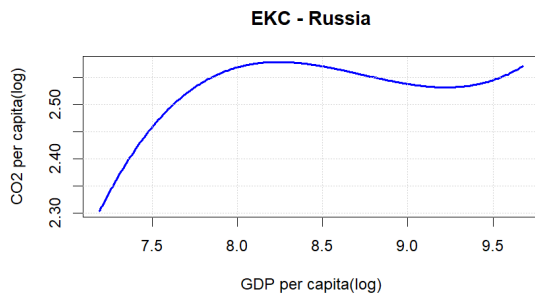


Table 9: Regression Coefficients – Russia

Coeff.	Estimate	Std. Err.	t-val	Pr(> t)
Intercept	-62.46724	29.85711	-2.092	0.0444
$\ln GDP$	22.47924	10.63481	2.114	0.0424
$\ln GDP^2$	-2.58224	1.25873	-2.051	0.0485
$\ln GDP^3$	0.09856	0.04950	1.991	0.0551

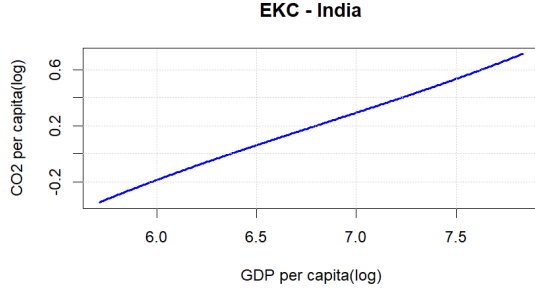


Table 10: Regression Coefficients – India

Coeff.	Estimate	Std. Err.	t-val	Pr(> t)
Intercept	-13.45948	17.35322	-0.776	0.444
$\ln GDP$	5.09131	7.75929	0.656	0.516
$\ln GDP^2$	-0.67903	1.15113	-0.590	0.559
$\ln GDP^3$	0.03320	0.05667	0.586	0.562

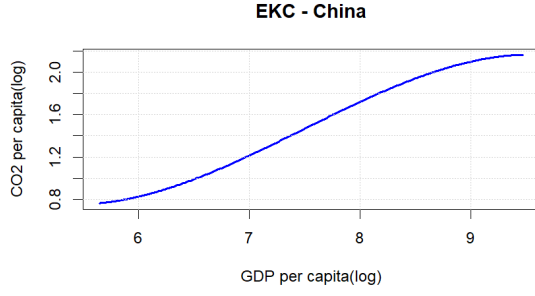


Table 11: Regression Coefficients – China

Coeff.	Estimate	Std. Err.	t-val	Pr(> t)
Intercept	14.821935	3.224365	4.597	6.39e-05
$\ln GDP$	-6.388302	1.316258	-4.853	3.04e-05
$\ln GDP^2$	0.922448	0.176724	5.220	1.05e-05
$\ln GDP^3$	-0.041079	0.007805	-5.263	9.25e-06

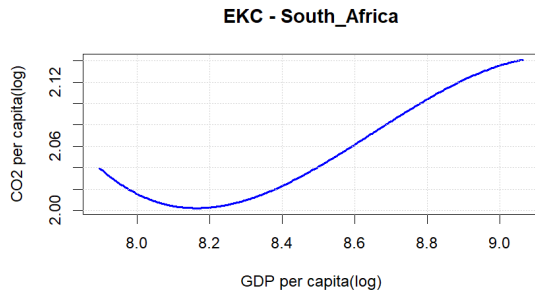


Table 12: Regression Coefficients – South Africa

Coeff.	Estimate	Std. Err.	t-val	Pr(> t)
Intercept	196.3670	249.6857	0.786	0.437
$\ln GDP$	-67.7999	88.5254	-0.766	0.449
$\ln GDP^2$	7.8610	10.4546	0.752	0.458
$\ln GDP^3$	-0.3028	0.4113	-0.736	0.467

By integrating the cube of per capita income into the polynomial function to explain the CO₂ per capita, we note that Russia's EKC first peaks, then declines, and then tends to rise again (effectively denoting an N-shape). For Brazil, India, and South Africa, the typical Environmental Kuznets Curve is not apparent here either, but only a linear and increasing relationship between CO₂ per capita and GDP per capita. China is a special case, as through a purely graphical analysis, we can see the initial (rising) phase of the EKC, which peaks and then tends to level off. This could demonstrate China's own intentions to peak carbon dioxide emissions in 2030 and then achieve carbon neutrality by 2060.

In 2024, the People's Republic of China accounted for approximately 17% of global GDP, with real GDP growth of 5.0%. This figure slightly exceeded the government's annual target and significantly outpaced the US growth rate of approximately 2.8%. In contrast to the monetary tightening observed in most major economies, the People's Bank of China pursued an expansionary monetary policy. Official interest rates were further reduced to reduce financing costs and stimulate private sector investment, particularly in strategic and high-growth sectors.

The energy sector remained a key driver of economic activity. China accounted for approximately one-third of total global clean energy investment, reflecting both state-led industrial policy and the continued expansion of domestic generation capacity. Clean energy investment contributed substantially to GDP growth, supported by concessional financing, production subsidies, and export incentives.

In particular, the "three new" high-tech sectors—photovoltaic solar cells, lithium-ion batteries, and electric vehicles recorded solid export performance. In 2023, exports in these segments increased 30% year-on-year, strengthening China's trade balance and cementing its position as a global leader in the production and distribution of green technologies.

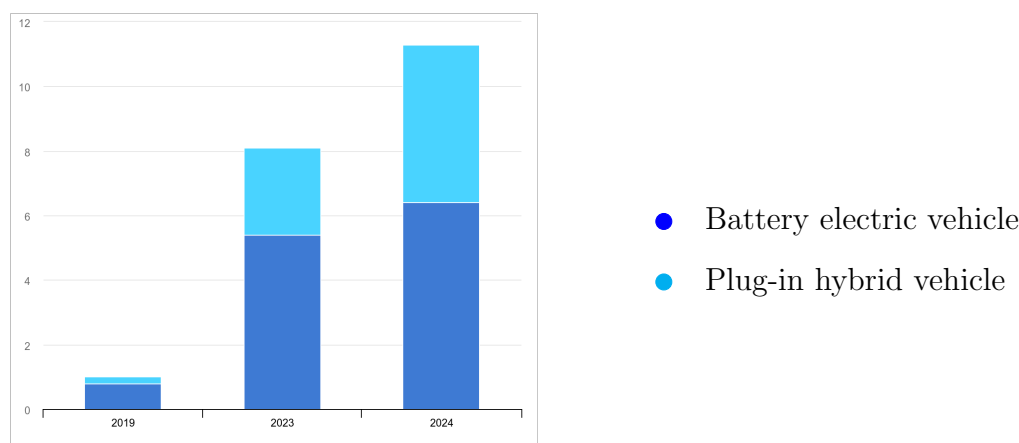


Figure 3: Electric car sales in China, 2019-2024

Despite rapid progress in renewable energy, China has continued to invest heavily in fossil fuel infrastructure, particularly coal-fired power generation. The country accounted for nearly all of the world's new coal-fired capacity additions in 2023-2024, citing concerns about energy security and grid stability. Coal remains the predominant source in China's energy mix, serving as a baseload resource to complement variable renewable generation.

When a country begins to grow rapidly, both in terms of production and population, the growth in demand is fueled by the demand for materials (cement and oil) and, of course, excessive energy demand. This is, however, the case for India, which in 2023 was the fastest-growing country in the world compared to the previous year. However, before focusing on decoupling carbon dioxide emissions from production, India, unlike other civilized and modern countries, must commit to increasing per capita income, healthcare, and therefore medical and health conditions, and, last but not least, life expectancy. From an environmental perspective, India issued so-called "green bonds" in 2023, which are intended to be used to finance green projects that benefit the environment (examples of green projects include the Rampur hydroelectric project).

However, all these policies are not able to reverse the curve in the immediate (short term), but they will probably be able to do so in the long term.

Even if the turning points are not very visible graphically, below I report a table

containing the levels of GDP per capita within which a reversal in the direction of CO₂ emissions occurs:

	Brazil	Russia	India	China	South Africa
GDP _{min} *	...	\$3788.08	...	\$230.03	\$3477.85
GDP _{max} *	...	\$10166.33	...	\$13795.60	\$9444.47

Table 13: Income was calculated using the formula on page 9. Since the discriminant for both Brazil and India is less than zero, the values have not been reported.

3.3 Conclusion

The definition of a tipping point in the EKC hypothesis is based on the concept of environmental awareness, which is highly correlated with social sustainability. This means that without social sustainability, a nation will never achieve environmental sustainability. Therefore, social indicators should be integrated into the EKC framework. For example, a country with a high literacy rate and low unemployment should have a lower level of environmental degradation than a country with a low literacy rate and high unemployment. Perhaps this is why developed countries have managed to reach the EKC tipping point, while developing and emerging economies have not yet reached it. This is a lesson that developing and emerging economies should learn from developed countries, rather than replicating their models in their own countries. To sustainably reach the tipping point, these economies should consider a people-led public-private partnership approach, which can ensure inclusive growth—a recipe for sustainable development. So far, we’ve only discussed emissions, production, and ways to reduce emissions, including through increased production. However, the assumption that economic growth automatically leads to a reduction in environmental impact appears weak, if not illusory. In this sense, a quote attributed to Brazilian activist Chico Mendes, a key figure in the fight to defend the Amazon and the rights of rural workers, is significant: “Environmentalism without class struggle is just gardening”. This, however, doesn’t mean we shouldn’t grow economically and stagnate in our current models; rather, it means seeking other alternatives, seeking other modes of production not aimed solely at unbridled profit, but more sustainable ones that protect all categories of people and workers. Consequently, the hope that the environmental curve will automatically reverse with an increase in GDP fails to explain the profoundly unequal and predatory structure of the dominant economic system. Without a systemic transformation that challenges the logic of profit and unlimited accumulation, any attempt at an ecological transition risks remaining superficial.

References

- [1] Grossman, G. M., & Krueger, A. B. (1995). *Economic Growth and the Environment*. The Quarterly Journal of Economics.
- [2] World Bank Open Data. <https://data.worldbank.org>
- [3] Fulufhelo Netswera (2025). *Making BRICS Work for its People and for those in the Global South*. Journal of BRICS studies.
- [4] O'Neill, J. (2001). *Building Better Global Economic BRICs*. Goldman Sachs.
- [5] Shahbaz, Muhammad and Sinha, Avik (2018). *Environmental Kuznets Curve for CO2 Emission: A Literature Survey*.
- [6] Luca Bettarelli, Davide Furceri, Prakash Loungani, Jonathan D. Ostry, and Loredana Pisano (2025). *Environmental Kuznets curve and green regulation*. United Nations University World Institute for Development Economics Research.
- [7] Yi Jiang, Tun Lin, and Juzhong Zhuang (2008). *Environmental Kuznets Curves in the People's Republic of China: Turning Points and Regional Differences*. ADB Economics Working Paper Series No. 141
- [8] IEA (2025), Electric car sales in China, 2019-2024. <https://www.iea.org/data-and-statistics/charts/electric-car-sales-in-china-2019-2024> *Licence: CC BY 4.0*